Log Anomaly Detection System: How to Run

G Sivakumar (23BCE1834), Afzal Ahamed A (23BCE1353). Guide: Manjula V, SCOPE, VIT Chennai

1. Requirements

Software	Version	Check with
Python	3.10 or newer	`python --version`
Node.js	16 or newer	`node --version`
MongoDB	7.0 or newer, on port 27017	`net start MongoDB` (as Administrator)
Ollama (optional, for the AI report)	any, with `llama3.2`	`ollama list`

Source code: https://github.com/sivakumar1306/Nosql-finalyear-project

2. One-time setup

Open PowerShell in the project folder and run:

```
pip install -r requirements.txt
cd log-dashboard
npm install --legacy-peer-deps
npm i ajv@8 --legacy-peer-deps
cd ..
ollama pull llama3.2
```

3. Start the system (two windows)

Start **MongoDB** and **Ollama** first (Ollama: Start menu, then check http://localhost:11434).

Window 1: backend (in the project folder):

```
python run_detection.py --generate
python run_backend.py
```

The first command loads 6 hours of demo traffic with 4 incidents and runs detection. Wait for `Uvicorn running on http://0.0.0.0:8000` and keep the window open.

Window 2: dashboard:

```
cd log-dashboard
npm start
```

Open	What it is
http://localhost:3000	Dashboard
http://localhost:8000/docs	API page (to inject logs)

4. Demo

1. **Dashboard:** pick a service in the sidebar to see its metrics, charts and anomalies.
2. **Inject failing logs:** on http://localhost:8000/docs open **POST /api/logs/batch**, click **Try it out**, paste 10 logs like the one below (same service), and click **Execute**. The response shows `"anomalies_detected": 1`.

   ```
   {"service_name":"payment-service","log_level":"ERROR","message":"Payment gateway timeout",

   "endpoint":"/api/payment","http_method":"POST","status_code":504,"response_time_ms":4200}
   ```

3. **Explain an alert:** refresh the dashboard and click the **eye icon** on the new anomaly to see the scores and the rules that fired.
4. **AI investigation:** click the **flask icon** (simulates a cascading failure), open the newest anomaly, then click **Investigate with AI**. Expected root cause: user-service.
5. **Model comparison (optional):** `python evaluation/model_comparison.py` (replaces the data).

5. Stop

Press **Ctrl+C** in both windows.

6. Common problems

Problem	Fix
`Failed to connect to MongoDB`	Run `net start MongoDB` as Administrator
Port 8000 or 3000 already in use	`netstat -ano | findstr :8000`, then `taskkill /PID <number> /F`
`No module named 'config'`	Start the backend with `python run_backend.py` from the project folder
`'react-scripts' is not recognized`	Repeat the one-time setup in `log-dashboard`
Dashboard shows zeros	Data is old: rerun `python run_detection.py --generate`, restart the backend
`"anomalies_detected": 0`	Same service tested within 5 minutes; wait or use another service
AI report unavailable	Start Ollama; check the model name with `ollama list`